Assignment

Dry Run

1. N-Queens Problem

Initial state: N = 4

board = $\begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}$

col = [0,0,0,0]
diag1 = [0 . . .]
diag2 = [0 . . .]

(i) Row 0
Place Q at (0,0)

(ii) Row 1
Try (1,2) - Place Q
Row 2 - No valid position - backtrack
Try (1,3) - place Q.

(iii) Row 2
Try (2,1) - place Q
Row 3 - no valid position - backtrack

(iv) Backtrack to Row 0
Place Q at (0,1)

(v) Row 1
Place Q at (1,3)

(vi) Row 2
Place Q at (2,0)

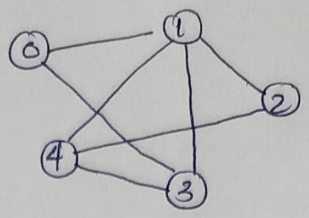
(vii) Row 3
Place Q at (3,2) $\longrightarrow$ Solution 1

similarly, find other solutions by placing Q at (0,2), and so on.

Valid solutions: $\begin{bmatrix} \cdot & Q & \cdot & \cdot \\ \cdot & \cdot & \cdot & Q \\ Q & \cdot & \cdot & \cdot \\ \cdot & \cdot & Q & \cdot \end{bmatrix}$ $\begin{bmatrix} \cdot & \cdot & Q & \cdot \\ Q & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & Q \\ \cdot & Q & \cdot & \cdot \end{bmatrix}$

Total solutions: 2

2. Hamiltonian Cycle



adj-matrix = $\begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}$

$V = 5$

path = [0 - - - -]

vis = [T F F F F]

(i) $u = 1$

neighbours of node 0 = [1, 3]

1 - edge exists - choose

path = [0 1 - - -] vis = [T T F F F]

(ii) $u = 2$

neighbours of node 1 = [0, 2, 3, 4]

2 → choose

path = [0 1 2 - -]

vis = [T T T F F]

(iii) $u = 3$

neighbours of node 2 = [1, 4]

4 → choose

path = [0 1 2 4 -]

vis = [T T T F T]

(iv) $u = 4$

neighbours of node 4 = [1, 2, 3]

3 → choose

path = [0 1 2 4 3]

vis = [T T T T T]

(v) $u = 5$

All vertices visited

check edge from last to first:

3 → 0 - exists

∴ Hamiltonian cycle found.

Hamiltonian Cycle: 0 1 2 4 3 0